Software Engineering

Candidate Preparation Guide

Software Development Engineer III

Congratulations on making it to the interview process for a Software Engineering role at OneByZero! This guide tells you what to brush up on ahead of each round. It does not list the specific questions we'll ask — the goal is to help you walk in well-prepared and confident, not to memorize answers.

The Interview Process at a Glance

Five rounds:

- R1 is a conversation with the Hiring Manager focused on fit, motivation, and communication (30–45 min).
- R2 is a technical screen covering coding and AWS (60 min).
- R3 is a live build session where you create a working AI-integrated application (60 min).
- R4 is a Culture & Behavioural round (60 min).

You may use whatever programming language and framework you're most comfortable with throughout the technical rounds.

General Tips — All Levels

- Know your own resume and past projects cold — be ready to walk through the 'why' behind decisions, not just the 'what'.
- Prepare 2–3 STAR-format stories (Situation, Task, Action, Result) covering a tough technical call, a disagreement or conflict, and a time you owned an outcome end-to-end.
- Practice narrating your thought process out loud while solving problems — interviewers are assessing how you think, not just your final answer.
- Revisit Big-O time and space complexity analysis so you can state it unprompted after solving a problem.
- Set up and test your preferred coding environment and language beforehand so setup doesn't eat into your time.
- Also configure debugging environment for your preferred language.
- Get comfortable using AI coding assistants as part of your normal workflow — using them well is expected and encouraged, not a shortcut to be hidden.
- Spend a little time understanding OneByZero's business — AWS-native, Agentic AI-focused systems integration for banks, telcos, insurers, and retailers across Asia Pacific — so you can speak to why the domain interests you.

SDE III — Preparing for Software Development Engineer III

Round 1 — Hiring Manager Fit Round

Be ready to talk about setting technical direction for a team or workstream, not just executing on someone else's design. Have an example of influencing a client or senior stakeholder, and a concrete story about growing another engineer's capability.

- Prepare an example where you set technical direction for a team or module, rather than just implementing a given design.
- Prepare a story about influencing a client or senior stakeholder's technical decision.
- Prepare a specific, detailed example of mentoring — what changed for the person you mentored, not just that you did it.
- Think of a time you held your ground under pushback from a client or stakeholder, and how you handled it.
- Reflect on your point of view on where the industry / GenAI space is heading, since you may be asked about it.

Round 2 — Competitive Programming + AWS Understanding

Data Structures & Algorithms — brush up on:

- Graph traversal / shortest-path style problems (e.g. BFS-based reasoning), including why an unweighted-shortest-path approach differs from a general search.
- Interval- or scheduling-style problems.
- Practice connecting an algorithmic solution to a real infrastructure concept — e.g. how a scheduling problem maps to connection pooling or concurrency management.
- Be ready to discuss how you'd approach a familiar problem if the input scaled by several orders of magnitude.

NOTE: IDE without autocompletion or any online interpreter with screenshare enabled

AWS — brush up on:

- Multi-region, multi-tenant architecture trade-offs — active-active vs. active-passive, and how you'd justify the choice.
- Designing for model/service failover across regions in a GenAI-serving platform.
- Resilience patterns in front of a third-party or managed AI service — circuit breakers, fallback/degraded modes.
- Cost governance at scale.
- Distributed tracing and observability concepts for diagnosing latency across a multi-stage pipeline.
- SLA-driven thinking — RTO/RPO concepts and how they shape architecture decisions.
- Compliance-oriented logging — what you'd need to capture and why, for audit purposes.
- Basic encryption/key-management concepts for protecting tenant data.

- Practice discussing auth, isolation, and throttling together as one coherent design, since they're usually asked as a connected scenario.
- AWS design principles, best practices.
- Serverless fundamentals.
- Levers that affect Lambda performance — memory allocation, package size, and how they relate to cost.
- Basic API Gateway concepts and how it fronts a serverless backend.
- CloudWatch basics — how you'd use logs to diagnose a performance issue.
- Load Balancing, EC2, IAM.

Round 3 — AI-Based Live Coding Session

- Multi-model pipeline design, specialist and an evaluation step before returning output.
- Cost and latency trade-offs across model tiers, and when a smaller vs. larger model makes sense.
- Guardrails — PII scrubbing, output validation, and content-policy checks.
- Fallback and error-recovery chains.
- Framing your design for enterprise deployment — auth, multi-tenancy, SLA compliance, governance, and observability built in from the start.

NOTE: Candidate should have an IDE, GenAI coding tool (like Claude, Codex, Cursor), API key for any AWS Bedrock / OpenAI / Anthropic API key

Round Example

R1 – Fit — Tell me about a time you changed the technical direction of a project after it was already underway.

R2 – DSA — Given a set of tasks with dependencies between them, determine a valid order to execute them in.

R2 – AWS — How would you design a system that must stay available even if an entire AWS region goes down?

R3 – AI Coding — Design a pipeline that routes a user query to the right specialized model and validates the output before it's returned.

Round 4 — Culture, Leadership & Behavioural Round

To prepare:

- Have specific, honest examples ready for how you handle disagreements, give and receive feedback, collaborate with others, and make decisions under ambiguity.
- Be prepared to go a level deeper on anything you've mentioned in your resume or earlier rounds—expect follow-up questions that explore your thinking and validate consistency.
- Keep your responses consistent across all interview stages. If your perspective has evolved or you simplified something earlier, be candid and explain it clearly.
- Think beyond describing experiences—be ready to explain *why* you made certain decisions, how you defended your technical viewpoint under constructive pushback, and what you learned.
- Have a genuine, well-thought-out answer for where you see yourself in the next few years and how this role aligns with your long-term goals.
- Reflect on the unique value you'd bring to the team and how you'd raise the bar through your technical strengths, leadership, or ways of working.
- Expect a light technical discussion woven naturally into the conversation. Treat it like any previous technical round rather than a separate assessment.
- Prepare thoughtful, senior-level questions for the leadership team—the quality of your questions often reflects your level of ownership, curiosity, and strategic thinking.

Final Notes Before Your Interview

- Get a good night's sleep — clear thinking matters more than last-minute cramming.
- Have a quiet, stable setup for your session, and test your camera/mic/internet beforehand if it's virtual.
- It's fine to ask clarifying questions before diving in — interviewers expect it and it's a positive signal, not a weakness.
- If you get stuck, talk through your thinking rather than going silent — partial credit comes from process, not just the final answer.
- Bring genuine questions for your interviewers — the quality of what you ask is itself part of what's being evaluated.

Learn More About OneByZero

Worth a look before your interviews — especially R1 and R5, where your motivation for joining and understanding of the business come up directly.

- Website: https://www.onebyzero.ai/
- Neo Platform: https://www.onebyzero.ai/neo
- Case Studies: https://www.onebyzero.ai/case-studies
- LinkedIn: https://www.linkedin.com/company/1bz/
- ASEAN Consulting Rising Star of the Year